MRI Reconstruction

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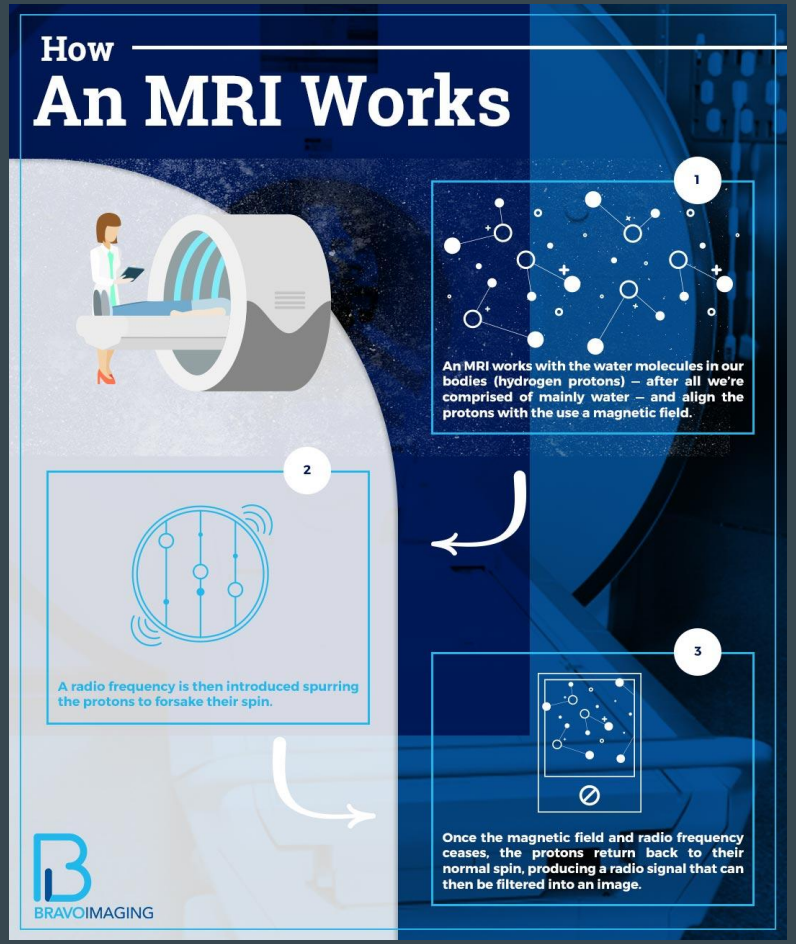
Background

MRI machines don't actually take pictures: they use magnets and the physical properties of water in our bodies to capture radio-frequency data.

Each radio signal is converted to a pixel in an image using a Fourier transform. By convention, we say that the MRI stores radio-frequency data in *k-space*, before being converted to an image.



Source: Bravo Imaging

The Problem

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Staying still for a long time is hard for

- Children
- Dementia Patients
- Patients with Huntington's, addictions, etc.

Shorter scan times = more accessible treatment

Potential Solution

What if, instead of measuring every value in k-space, we only directly measured a subset of them?

- Quicker scans, but more blurry images

Can we can **reconstruct** a partially formed MRI image using optimization?

- Quick scans with more detailed images

Mathematics

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

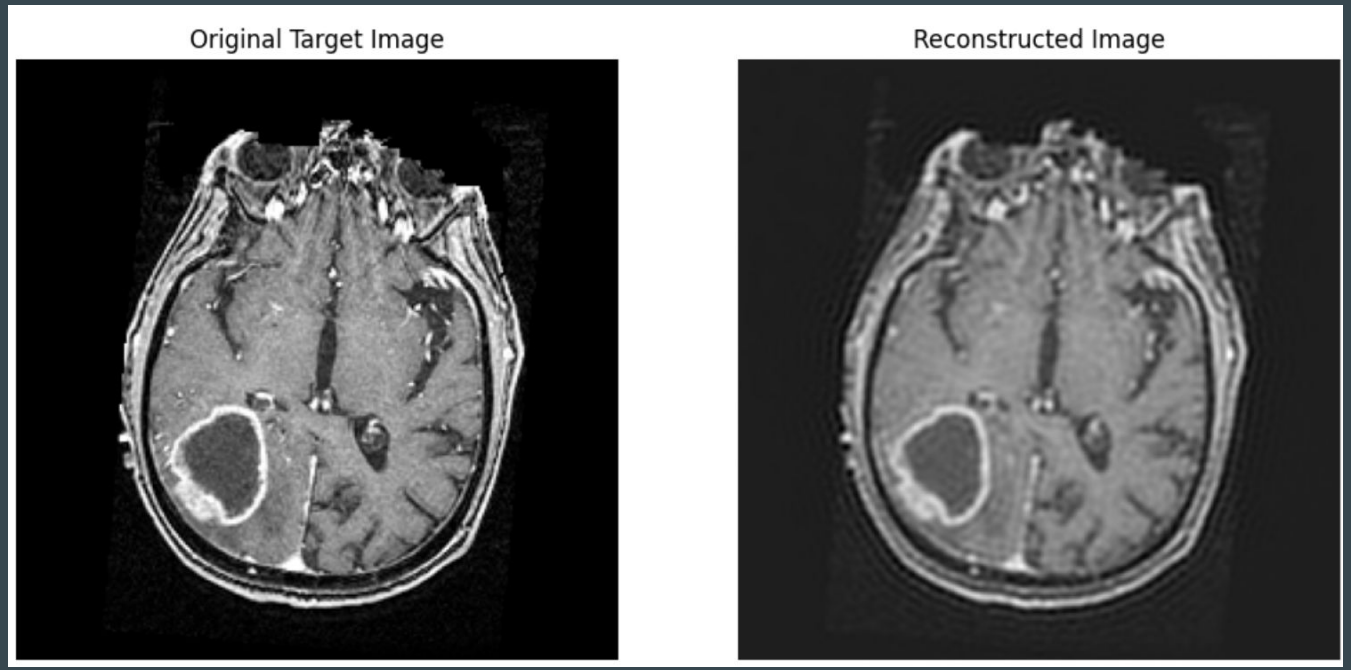
$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We use **Gaussian kernels** to impose smoothness.

Implementation

- **Data** : UPenn-GBM Cancer Imaging Archive. Selected target image.
  - Create kernel basis & lots of data preparation
- **Simulate** the under-sampling of frequencies: 'masking' the data by adding a filter to only show low-frequency waves.
- **Solve** : Using an Adam (Adaptive Moment Estimation) optimizer.
- **Validate** : (1) compare reconstructed image to original MRI target image and (2) measure loss over iterations.

Results



Pro:

- Working prototype w/real data

Con:

- Blurry image
- No frame of reference for accuracy

Next steps

- **Improve Validation:** How good is our model vs. if we filled all empty pixels with blank space? We run a simulation with zero-filled FFT to simulate the null case.
  - Provides context for our reconstruction quality
- **Test with different kernels:** LaPlace kernel allows for sharper lines, which may create a more accurate image.